

Transparent Polymer-Ceramic Hybrid Antifouling Coating with Superior Mechanical Properties

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Antifouling coatings are often required to have both excellent fouling resistance and mechanical properties for their applications. This study reports a water-borne antifouling coating consisting of an amine-terminated hyperbranched polysiloxane and a fouling resistant epoxy-zirconium particle, where the former is prepared using amine-functionalized silanes, and the latter is prepared via the reaction of zirconium alkoxide with epoxy-functionalized silanes and zwitterionic silanes. Such a hybrid coating is transparent (>99.5% transmittance) and exhibits a combination of polymeric and ceramic characteristics, namely, it has not only high hardness (7–9 H) and adhesion (≈ 3 MPa) to substrates but also good flexibility (≤ 10 mm bending diameter). The presence of the zwitterionic group allows the coating to have excellent oil repellency and fouling resistance. The effects of composition and structure on the mechanical properties and antifouling performance are investigated. This study aims to develop antifouling coatings to be used in foldable displays, optical sensors, and biomedical facilities.

precursors such as functionalized silanes or metal alkoxides can have not only outstanding antifouling performance but also good mechanical properties,^[9] although their mechanical properties are inferior to those of glass or ceramic. To meet the requirements for applying to foldable displays and public facilities, their hardness, scratch resistance, adhesion to substrates, and flexibility should be improved.^[10] Recently, flexible hard hybrid coatings have been developed for foldable displays, which exhibit superior mechanical properties but lack antismudge or antimicrobial functions.^[11] Furthermore, some omni-phobic flexible hard hybrid coatings have been reported.^[12] The coatings exhibit outstanding antismudge ability besides their superior mechanical properties. Yet, their excellent antismudge ability mainly

1. Introduction

The attachment of micro-organisms onto surfaces generally results in biofouling that can cause serious problems in biomedical, marine, and other applications and even spread diseases.^[1] Conventional antifouling systems usually containing toxic metals or organic compounds have proven to have a negative ecological impact. Today, eco-friendly yet high antifouling performance antifouling coatings are expected.^[2] Accordingly, siloxane-based fouling-release coatings,^[3] antifouling copolymers,^[4] and polymer brushes^[5] have been prepared to inhibit biofouling. However, these polymer-based materials often exhibit poor mechanical properties, and they are readily destroyed and even peeled from the substrate.^[6]

Actually, some effort has been made to investigate alternatives to polymer-based counterparts.^[7] One effective approach is to develop organic–inorganic hybrid antifouling coatings by sol–gel chemistry.^[8] Such coatings prepared from various

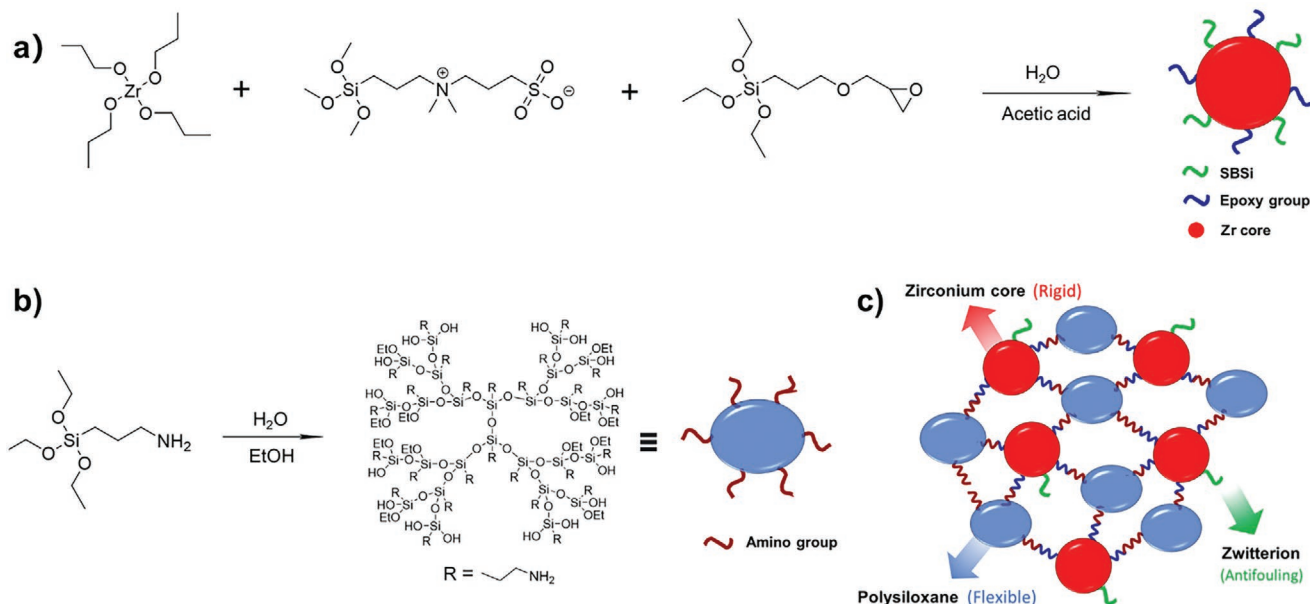
comes from polydimethylsiloxane (PDMS), while PDMS has proven to show limited micro-organism resistance ability.^[13] Without antimicrobial property, the surface of coating would harbor micro-organisms, which would probably increase the risk of being infected with diseases.^[14] Besides, the coatings usually contain volatile organic compounds which may pollute the environment during their use. Therefore, it is essential to develop environmentally-friendly antifouling coatings with superior mechanical properties.

Zirconium has been used to improve the mechanical properties of ceramic or glass because of the formation of Zr–O–Si structure.^[15] Given that zirconium alkoxide is water-soluble, the resultant coating is water-borne and hence eco-friendly.^[11b] In the present study, we synthesized a highly cross-linked zwitterionic epoxy–zirconium particle (ZPx) using (3-glycidyloxypropyl)trimethoxysilane (KH560), tetrapropyl zirconate (TPOZ), and sulfobetaine silane (SBSi). ZPx particles react with a hyperbranched siloxane (HPSi) prepared from 3-amino-propyltriethoxysilane (KH550), yielding a water-borne, and highly transparent coating. Since ZPx is covalently bonded with HPSi, the coating demonstrates not only good flexibility but also high hardness as well as good adhesion to substrates. Furthermore, the presence of zwitterionic group allows the coating to possess oil repellency and antibacterial ability. We have examined the antifouling performance and mechanical properties. We attempt to develop an eco-friendly and high-performance antifouling coating.

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Scheme 1. a) Preparation of the epoxy–zirconium particle. b) Preparation of the amine-terminated hyperbranched polysiloxane. c) Schematic representation of the cross-linked network of hybrid coating.

2. Results and Discussions

Scheme 1 illustrates the preparation of the epoxy-zirconium particle using tetrapropyl zirconate (TPOZ), 3-glycidyloxypropyltrimethoxysilane (KH560), and sulfobetaine silane (SBSi), whereas the synthesis of an amine-terminated hyperbranched polysiloxane using 3-aminopropyltriethoxysilane (KH550) is also presented. For convenience, the former is designated as ZPx and the latter as HPSi, where x is the mass ratio of SBSi in the TPOZ and KH560. ZPx reacted with the HPSi to yield a cross-linked network, that is, the zirconium–siloxane hybrid coating designated as HP-ZPx. Hyperbranched polysiloxane has proven more effectively to improve the mechanical properties of coatings than the corresponding linear one because it acts as a crossing linker and has more interacting or reacting sites with other components.^[16] To extract the role of the hyperbranched structure, KH550-ZPx samples were prepared following the same protocol for HP-ZPx. The successful synthesis of SBSi and HPSi was confirmed via nuclear magnetic resonance spectroscopy (Figures S1–S3, Supporting Information).

In the following experiments, the coating was deposited on glass slide, ceramic, steel, or polyethylene terephthalate (PET) by using the drop-casting method. A tetrahedral wet film fabricator was used to control the thickness of the coating, which was measured by using a coating thickness gauge (Phynix, Model Surfex).

Adhesion strength is a critical parameter for any given coating.^[17] We measured the strength of the coating adhered to the steel, glass, ceramic, or PET. **Figure 1** shows that the coating has strong adhesion to the steel, the glass, and the ceramic. This is probably because the hybrid coating is covalently bonded to the substrates with polar groups such as hydroxyl groups.^[18] KH550-ZP0 has a lower adhesion to steel (3.41 ± 0.22 MPa), glass (3.29 ± 0.15 MPa), or ceramic (3.14 ± 0.10 MPa) than

HP-ZP0 (3.72 ± 0.20 , 3.61 ± 0.17 , and 3.21 ± 0.10 MPa, correspondingly), indicating that the hyperbranched polysiloxane can improve the adhesion. As the content of zwitterionic silane increases, the adhesion strength decreases, and HP-ZP15 has the lowest adhesion to steel (2.76 ± 0.26 MPa), glass (2.97 ± 0.11 MPa), and ceramic (2.37 ± 0.09 MPa). Nonetheless, the coatings show superior adhesion to steel, glass, and ceramics substrates. Note that the coating has low adhesion to PET (≈ 0.9 MPa). This is understandable because PET surface is too hydrophobic.^[19] Actually, for previous antifouling coatings with introduced PEG or zwitterions tested by the same method, they generally exhibit adhesion strength values of

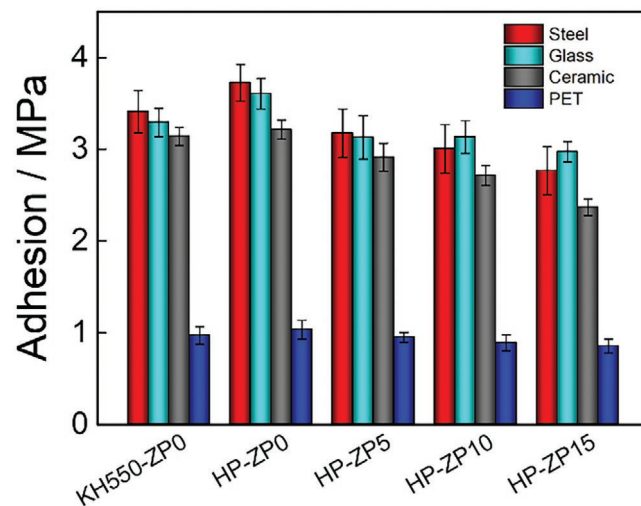


Figure 1. Adhesion of each hybrid coating to steel, glass, ceramic, and PET

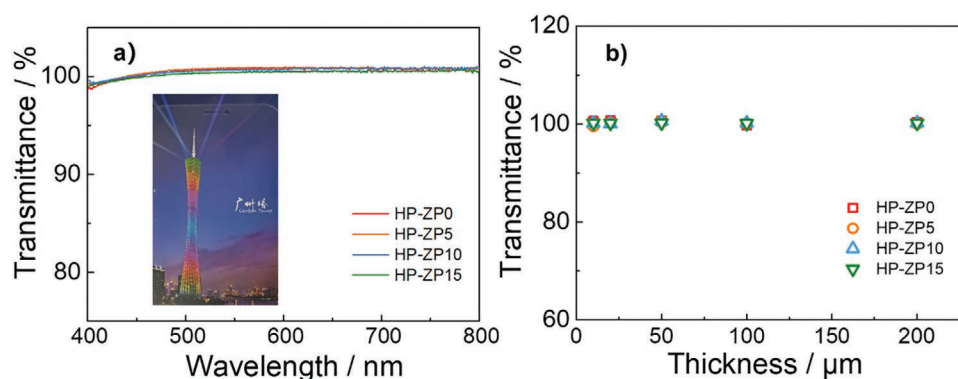


Figure 2. a) Transmittance spectra of the hybrid coatings with a thickness of $\approx 20 \mu\text{m}$. b) Thickness dependence of the average transmittance of the hybrid coatings.

below 2.5 MPa on glass fiber enforced epoxy panels or steel,^[20] which are inferior than the worst sample HP-ZP15.

Figure 2 shows the transmittance spectrum of each coating on a glass substrate, where the transmittance of the glass substrate was used as the baseline. Each coating sample with a thickness of $\approx 20 \mu\text{m}$ exhibits similar spectrum, that is, they have $>98.5\%$ transmittance at any wavelength and $>99.5\%$ average transmittance from 400 to 800 nm, indicating their good transparency. Actually, either the coating film or solution is homogeneous. This is because the size of the particles in the coating is smaller than the visible light wavelength.^[21] When the coating was applied to the screen of a smartphone, the optical clarity does not decrease (The inset in Figure 2a). Figure 2b shows the average transmittance of each coating with the thickness ranging from 10 to 200 μm . Each HP-ZPx coating with a thickness of 10 μm shows a transmittance of $\approx 100\%$, indicating that they have a transmittance close to the glass substrate. Particularly, the average transmittance does not vary even as the thickness increases up to $\approx 200 \mu\text{m}$.

Mechanical properties determine the durability of coatings applied to foldable displays. Therefore, we measured the mechanical properties of hybrid coatings, where each coating with a thickness of $\approx 10 \mu\text{m}$ was coated on glass. We first measured the pencil hardness of each coating (Table S1, Supporting Information). HP-ZP0 possesses the highest pencil hardness (9 H). The introduction of zwitterionic silane would decrease the hardness. Here, the pencil hardness of HP-ZP5, HP-ZP10, and HP-ZP15 is 8, 7, and 5 H, respectively. The quantitative hardness and elastic moduli data of the hybrid coatings were measured by using a nanoindenter. Figure 3a shows that KH550-ZP0 has a hardness of $209 \pm 2 \text{ MPa}$, lower than HP-ZP0 ($224 \pm 4 \text{ MPa}$), indicating that hyperbranched polysiloxane can more effectively improve the hardness than KH550. However, the hardness of HP-ZPx decreases as the content of the zwitterionic groups increases. HP-ZP5, HP-ZP10, and HP-ZP15 have a hardness of 186 ± 4 , 170 ± 1 , and $159 \pm 1 \text{ MPa}$, respectively. This is understandable because zwitterionic groups serve as a plasticizer and reduce the hardness. The hardness in this system seems to show lower value than some of the previous counterparts, even though the coatings have the same or higher pencil hardness, which is probably because of the different experimental methods used.^[11,12] Actually, there is no direct correspondence between the value of pencil hardness and that of

hardness obtained from nano-indentation since the hardness measured from nano-indentation changes depending on the substrates, coating thickness, and penetration depth.^[9c,22]

Figure 3b shows the load-displacement curve determined by the nanoindenter. The tip penetrated the same depth of 500 nm into each coating, and the harder coating requires more loading force to damage its surface. Clearly, this result is in accordance with the pencil hardness test or the nano-indentation test and indicates HP-ZP0 is the hardest and the hardness of HP-ZPx decreases with zwitterionic silane content. For the same reason, the moduli of the coatings decrease with the content of zwitterionic silane (Figure 3c).

The wear resistance of each coating was determined using a wear tester. The coating was first abraded using steel wool with each circle in Figure 3d denoting a cycle of one forward and one backward movement of the steel wool. After the coating was abraded for 2 cycles, numerous scratches were observed on PET surface. However, this phenomenon did not happen in the case of HP-ZPx samples, indicating that they have a higher wear resistance than PET. Five cycles abrasion made no difference among HP-ZP0, HP-ZP5, and HP-ZP10 samples, whereas numerous scratches were observed on HP-ZP15 surface due to its low hardness. After 10 cycle abrasion, HP-ZP5 and HP-ZP10 surfaces were also marked by numerous scratches. However, HP-ZP0 was able to bear 20 scratching cycles without visible scratches, so it has the best wear-resistant ability, and the wear resistance decreases with zwitterionic silane content. Anyway, HP-ZP0, HP-ZP5, and HP-ZP10 exhibit enough wear resistance and can meet the general requirements.

We examined the flexibility of the hybrid coatings by using bending (Figure 4a) and rolling-up tests (Figure 4b), where each coating with a thickness of $\approx 10 \mu\text{m}$ was coated on PET. As shown in Figure 4a, after 5 bending cycles, numerous cracks appeared on the surfaces of KH550-ZPx, indicating their brittleness. In contrast, no crack was observed on HP-ZPx surfaces even after 20 bending circles, implying that the introduction of hyperbranched silanes can effectively improve the flexibility of hybrid coatings. We chose KH550-ZP0 and HP-ZP0 with high hardness and high moduli to further examine the flexibility, where they were rolled up into a cylinder with a diameter of $\approx 10 \text{ mm}$. Figure 4b shows that KH550-ZP0 film severely cracked following one rolling-up cycle due to its brittleness, but no crack was observed on HP-ZP0 surface even after

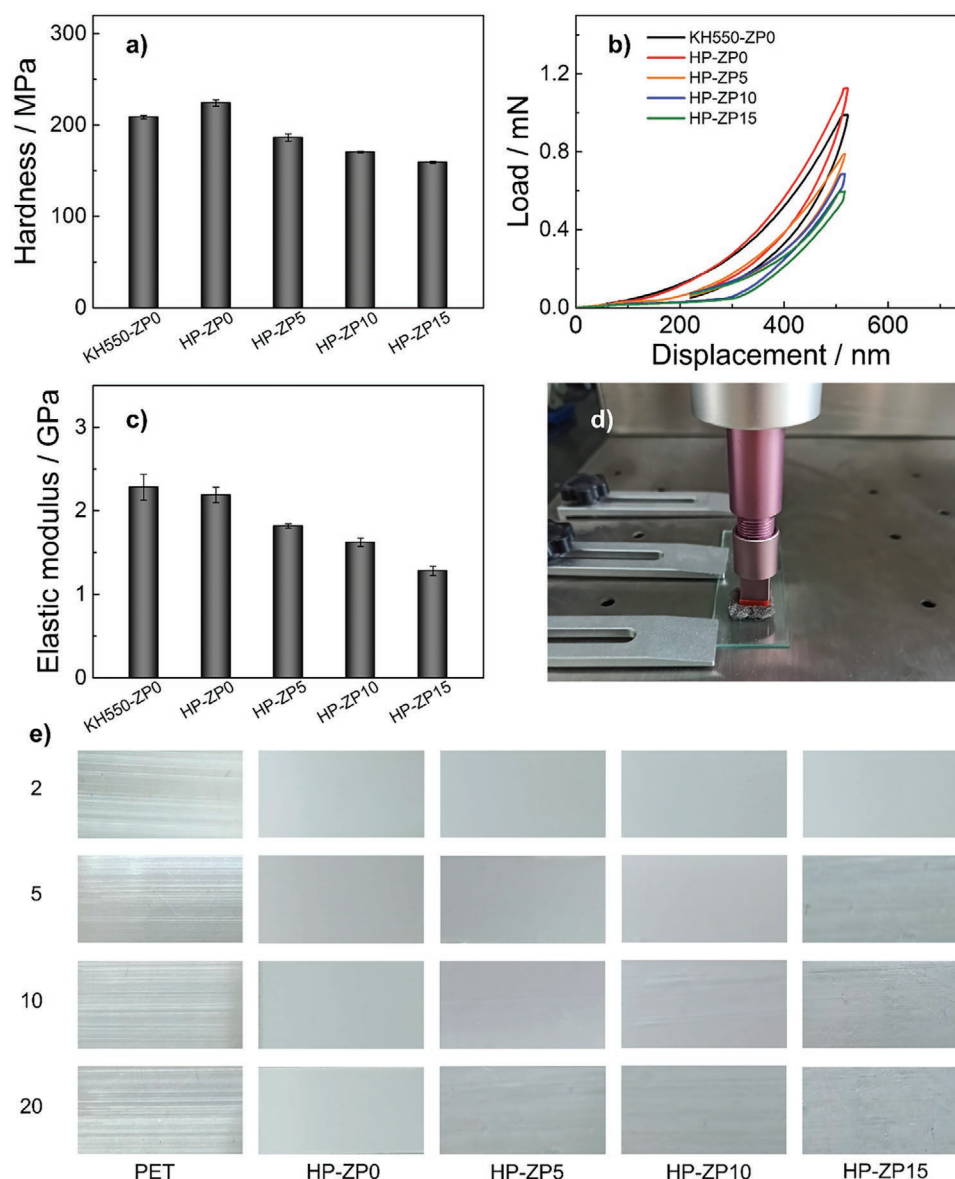


Figure 3. a) Hardness, b) load–displacement curve, and c) elastic modulus of each hybrid coating. d) Photograph of the wear tester. e) Photograph of PET and HP-ZPx abraded using steel wool for 2, 5, 10, and 20 cycles, respectively.

10 rolling-up circles. The results suggest the hyperbranched polysiloxane can significantly improve the flexibility of coatings. This is because the hyperbranched structure acting as cross-linker allows the polysiloxane to have more interaction sites with zirconium particles so that the coating can undergo more elastic deformation. It is known that hard but flexible materials generally exhibit a hardness-to-elastic modulus ratio (H/E ratio) exceeding 10%.^[11,12] Here, the H/E ratio of KH550-ZP0 is $\approx 9\%$, whereas that of each HP-ZPx is over 10%. The facts indicate HP-ZPx can store the work done by indenter as elastic energy and recover from the damage later. That is why HP-ZPx exhibits flexibility.

The flexibility of coating film was quantitatively measured by flexibility tests (Table S1, Supporting Information). HP-ZP0 is flexible since it has a diameter as small as 10 mm. As

zwitterionic silane content increases, the flexibility increases. For HP-ZP5, HP-ZP10, and HP-ZP15, the diameters are 5, 3, and 1 mm, respectively. Therefore, HP-ZPx has enough flexibility to find applications in foldable displays and other fields.

Figure 5a shows the static and dynamic water contact angles (WCA). HP-ZP0 has a static WCA of $\approx 83.4^\circ$. As the zwitterionic silane content increases, the static WCA of HP-ZPx decreases in that the surface becomes more hydrophilic. HP-ZP5, HP-ZP10, and HP-ZP15 have static WCAs of $\approx 80.4^\circ$, $\approx 78.5^\circ$, and $\approx 76.0^\circ$, respectively. For dynamic WCAs, only a small difference can be observed in advancing WCAs, from HP-ZP0 ($\approx 88.1^\circ$) to HP-ZP15 ($\approx 78.1^\circ$). This is because the advancing WCA reflects the hydrophobicity of the coatings. The receding WCA of each coating demonstrates a relatively significant variation since it is related to the hydrophilicity.^[23] HP-ZP0 has a receding WCA

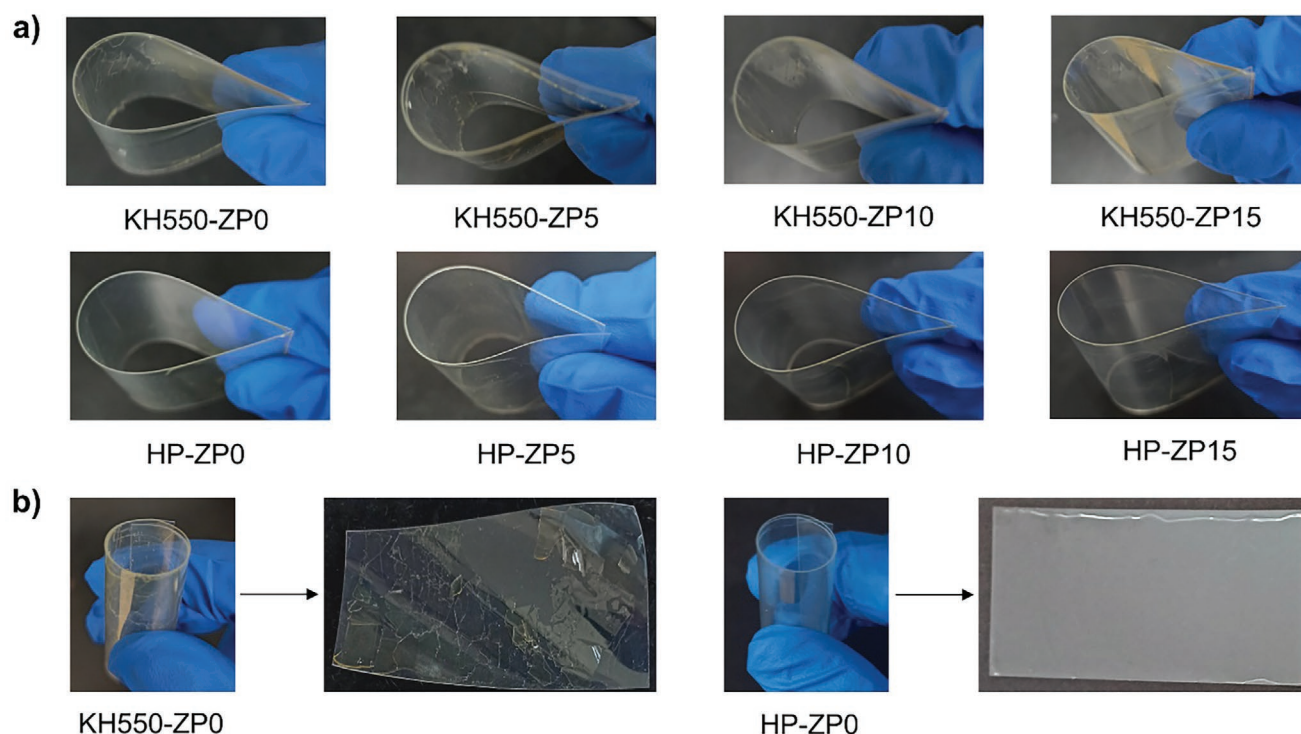


Figure 4. Photographs of the hybrid coatings in a) bending test and b) rolling-up test.

of $\approx 45.4^\circ$, but it significantly decreases to $\approx 21.3^\circ$ (HP-ZP15) as the zwitterionic silane content increases. The results indicate that zwitterions in this system can migrate to the surface of coatings and hence increase the hydrophilicity and decrease the

WCAs of them. We measured the underwater oil contact angle (UWOCA) of each coating to determine its oleophobic property.^[24] HP-ZP0 has a UWOCA of $\approx 46.0^\circ$, indicating that it is oleophilic. As the content of zwitterionic silane increases, the

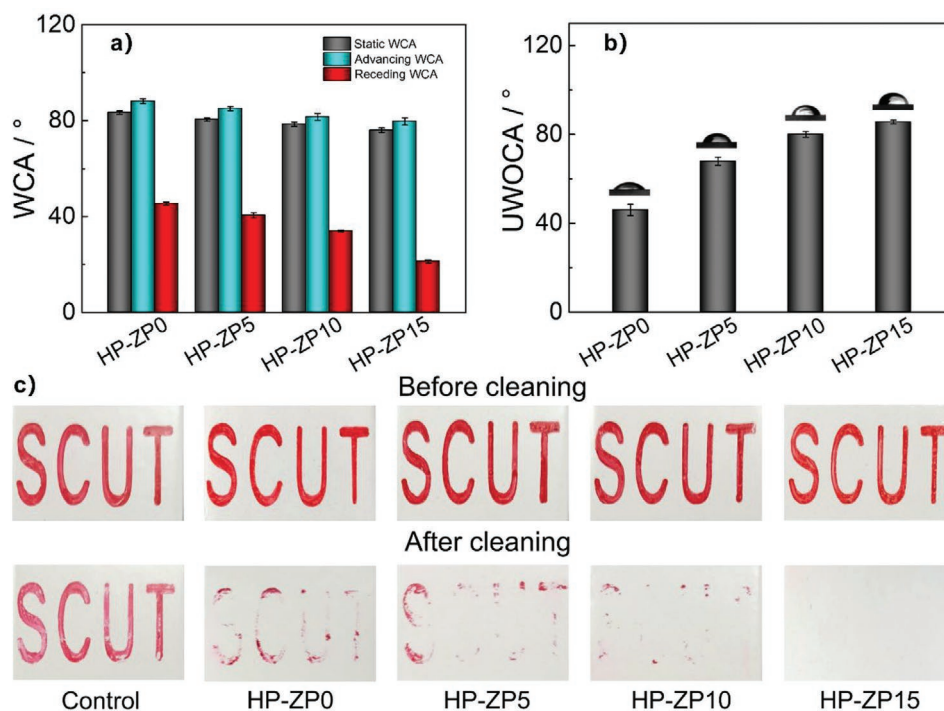


Figure 5. a) WCA, b) UWOCA, and c) oil repellency of each hybrid coating.

UWOCA of HP-ZP15 increases to $\approx 85.6^\circ$ by $\approx 39.6^\circ$, implying the presence of zwitterionic groups increases the oleophobicity.

The oil-repellent ability of each coating was determined with a ceramic plate as the control (bare substrate). As shown in Figure 5c, the control and ceramic plate coated with the coating were inscribed by using an oily pen. The coating was then immersed in deionized water for 4 h and gently cleaned with a piece of tissue. Following the cleaning, the writing on the control could barely be erased since ceramics do not have oil-repellent ability. The writing on HP-ZP0 was somewhat erased probably because of the presence of polysiloxane. However, as zwitterion content increases, the oil repellence increases, and the writing on HP-ZP15 was completely erased.

Chemical stability of the coatings is important for their applications. We then conducted tests of resistance to UV radiation and liquids for 24 h, with HP-ZP0 and HP-ZP10 as the typical samples to be tested. We used static WCAs and pencil hardness of coatings to evaluate their chemical resistance performance. The coatings were deposited on a glass slide, where WCA of the substrate is $\approx 41.3^\circ$. Before the test, WCAs of HP-ZP0 and HP-ZP10 are 83.4° and 78.5° , and hardness values of them are 9 and 7 H, respectively. As shown in Table S2, Supporting Information, after being exposed to UV radiation for 24 h, HP-ZP0 and HP-ZP10 have WCAs of $\approx 84.4^\circ$ and $\approx 79.6^\circ$, respectively, similar to their original values, and their hardness is 9 H and 7 H, respectively, the same as before. Besides, we tested their transmittance spectra (Figure S4, Supporting Information), which showed they kept superior optical clarity. In conclusion, the coatings have excellent resistance to UV radiation.

We also selected nonpolar organic solvents (hexane, xylene), polar organic solvents (ethanol, DMSO), and acidic or basic aqueous solution (0.1 M HCl aqueous solution, 0.1 M NaOH aqueous solution) for liquids resistance tests. The results show all organic solvents did not influence the pencil hardness of our coatings and the variation of WCAs is slight, implying the coatings are greatly inert to them. However, HP-ZP0 and HP-ZP10 show a great variation in WCAs and pencil hardness values for acidic and basic solution, indicating they have limited acid resistance and base resistance in this test. Yet, one should note immersing in 0.1 M HCl or 0.1 M NaOH aqueous solution for 24 h is a much harsher condition than the actual environment. For simulating the real scenario, we dropped these two solutions on the surface of our coatings and adsorbed them with

tissue paper after 10 s, and coatings kept intact more than 50 repeating times. In short, the results show our coatings exhibit excellent resistance to UV radiation and organic solvents and enough resistance to acidic or basic solution in simulated real conditions.

It is well-known that PEG and zwitterionic groups can form hydration layer with its interfacial water molecule, which serves as a barrier to effectively resist bacteria. However, PEG has been reported to be susceptible to oxidation reaction and loss its efficacy.^[25] Therefore, we introduced zwitterions into this system to improve its resistance to bacteria. In the wettability measurement, we proved zwitterions in the epoxy-zirconium particle can migrate to the surface of coatings, which will endow them with antibacterial ability. To investigate the antibacterial ability of the hybrid coatings, we conducted antibacterial assays where a cleaned glass slide was used as the control (bare substrate) and the average relative bacteria adhesion value of the control was regarded as 100%. We selected *Pseudomonas* sp. (*P. sp.*), *Escherichia coli* (*E. coli*), and *Staphylococcus aureus* (*S. aureus*) as models since they widely exist in marine environments or our daily environments. Figure 6 shows a large number of bacteria *P. sp.* were on the surfaces of the control and HP-ZP0. As the zwitterionic silane content increases, the bacteria on HP-ZPx surface decrease, and almost no bacteria were observed on HP-ZP15 surface ($\approx 6.5\%$ bacteria adhesion). Clearly, the zwitterionic groups allow the hybrid coatings to have antibacterial ability. The same phenomenon was observed for *E. coli* and *S. aureus*. As the zwitterion content increases, the adhesion of bacteria decreases, or the fouling resistance increases.

Compared with PEG or zwitterion-modified polymer-based and hybrid antifouling systems,^[3–9,20] the results show our material exhibits similarly excellent antifouling performance but much higher hardness, modulus, and adhesion strength. For hybrid coatings designed for foldable displays, previous Zr containing materials or flexible hard hybrid coatings exhibit superior mechanical properties but with no antimicrobial or antimicrobial functions.^[11] Recently, GPOSS-like flexible hard hybrid coatings with excellent antismudge ability have been developed.^[12] In comparison with GPOSS-like coatings, scratch resistance of our material is somewhat inferior than them. However, HP-ZPx outperforms in their flexibility, that is, HP-ZPx can not only be bent into a closed U cross-section but also be rolled up into a cylinder without cracking. On the other

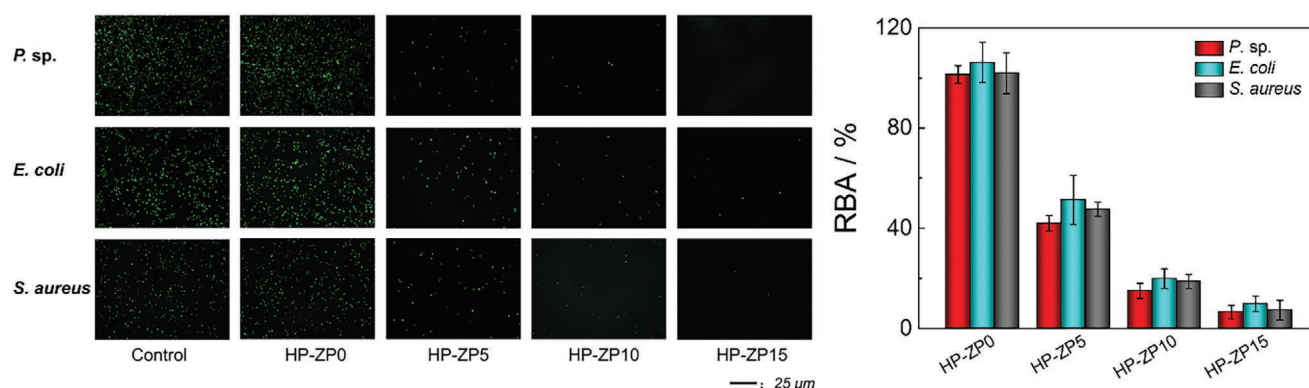


Figure 6. Fluorescence microscopy images of *P. sp.*, *E. coli*, and *S. aureus* on the surface of hybrid coatings and the relative bacteria adhesion (BRA).

hand, even though HP-ZPx exhibits inferior oil repellency than GPOSS-like coatings, they own excellent anti-bacterial property, which can prevent the surface of screen from being the harbor of micro-organisms. In conclusion, the developed transparent coating in this study exhibits a combination of ceramic-like hardness, adhesion to substrates, wear-resistance, and polymer-like flexibility. Besides, the coating simultaneously has oil repellency and antimicrobial ability toward diverse bacteria.

3. Conclusion

We have developed a hybrid antifouling coating by using an epoxy-zirconium particle and an amine-terminated hyperbranched polysiloxane. The coating is water-borne and highly transparent. It exhibits superior mechanical properties and excellent antifouling performance. The hyperbranched polysiloxane plays an important role in improving the coating's adhesion to substrates and its hardness as well as its flexibility. The presence of zwitterionic silanes allows the coating to have oleophobicity and antibacterial ability. The coatings are expected to find application in foldable displays or other fields.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

antifouling, hyperbranched polymers, polymer-ceramic hybrid, water-borne, zwitterions

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